Tuning SiO2 core size

"How can the SiO2 core size be tuned in mesoporous core-shell particles?" *Generated by LeapSpace*

The **SiO_2 core size in mesoporous core-shell particles is tuned mainly by controlling Stöber/sol-gel growth conditions**—especially **TEOS amount, water content, ammonia or other base catalyst, ethanol/solvent composition, mixing mode, reaction time, and temperature**—which together shift the core from smaller, monodisperse spheres to larger or more broadly distributed particles [70] [71] [72] [73] [74] [75].

Main tuning levers

Variable	Effect on SiO_2 core size / distribution	What the evidence shows
TEOS concentration / feed rate	Increasing precursor availability generally changes particle growth and final size [73] [75] [76]	TEOS feed rate and concentration are identified as key controls in Stöber-derived silica size tuning [73] [75] [77]
Water content	Can move the system between aggregated, discrete, and size-controlled regimes [70] [75]	In the Stöber process, discrete particle size is dictated by water content within the classical regime [70]
Ammonia / base catalyst	Changes hydrolysis-condensation rates and thus particle growth [72] [73] [74]	Higher or different catalysts altered average size, BET area, and pore volume in mesoporous silica [72]
Ethanol / solvent composition	Alters particle diameter and size distribution [71] [78] [79] [80]	A solvent-varying Stöber method produced silica spheres from **70 to 400 nm** by changing ethanol volume [71]
Surfactant / template concentration	Affects mesoporosity and can also shift size and morphology [75] [79] [81]	Increasing CTAB raised pore volume and surface area while decreasing pore diameter and particle size in mesoporous silica [79]
Mixing mode and duration	Impacts homogeneity and size distribution [73]	Better stirring/bidirectional mixing improved uniformity and reduced multimodal distributions [73]
Reaction time	Can narrow distribution and increase textural development [82]	Longer reaction time increased pore volume and surface area while maintaining similar shell thickness in one mesoporous core-shell silica system [82]
Temperature	Affects final diameter and distribution [70] [74] [75] [83]	Temperature is explicitly listed as a size-controlling variable in Stöber silica synthesis [74] [75]

Practical takeaway

If your goal is to tune the **SiO_2 core** for a mesoporous core-shell particle, the most reliable approach is to **use a Stöber-type silica core synthesis and adjust one variable at a time**: **TEOS, water, ammonia/base, ethanol, and mixing** are the strongest size controls, while **surfactant/template conditions** mainly shape the mesoporous shell but can also influence overall particle size and uniformity [70] [71] [72] [73] [74] [75].

Bottom line: the SiO_2 core size is not usually tuned by a single "size knob"; it is controlled by the **hydrolysis–condensation balance and nucleation/growth kinetics** set by solvent, water, base, and precursor feed, with surfactant and mixing refining the final size distribution and mesostructure [70] [72] [73] [74] [75].

Confidence level: **High —** multiple independent Stöber/mesoporous silica studies consistently identify the same synthesis parameters as the main size controls, and the direction of their effects is directly supported by the retrieved evidence [70] [71] [72] [73] [74] [75].

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